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Skill Builder : 3

```
In [1]: pixel_a = 50
pixel_b = 150
threshold = 50
diff = abs(pixel_b - pixel_a)
edge_type = "Strong edge" if diff >= threshold else "Weak edge"

print(f"Pixel A = {pixel_a}")
print(f"Pixel B = {pixel_b}")
print(f"Intensity Difference = {diff}")
print(edge_type)
```

Pixel A = 50
Pixel B = 150
Intensity Difference = 100
Strong edge

```
In [2]: import cv2
import numpy as np
import matplotlib.pyplot as plt

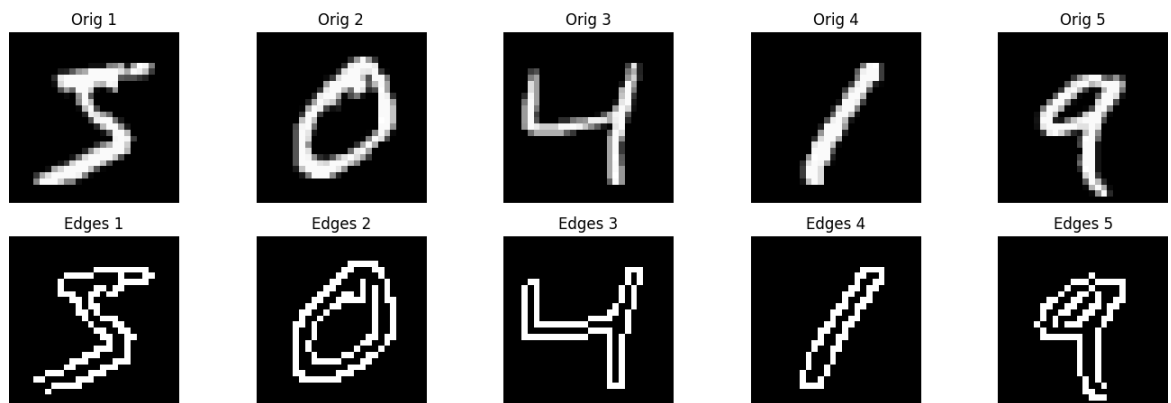
data = np.load(r"C:\Users\HP\OneDrive\Desktop\CV&DL\student_micro_dataset2.npz")
mnist_x = data['mnist']
cifar_x = data['cifar']

fig, axes = plt.subplots(2, 5, figsize=(14, 5))
fig.suptitle("MNIST | Top: Original Bottom: Canny Edges", fontsize=12)
for i in range(5):
    edges = cv2.Canny(mnist_x[i], 50, 150)
    axes[0, i].imshow(mnist_x[i], cmap='gray')
    axes[0, i].set_title(f"Orig {i+1}")
    axes[0, i].axis('off')
    axes[1, i].imshow(edges, cmap='gray')
    axes[1, i].set_title(f"Edges {i+1}")
    axes[1, i].axis('off')
plt.tight_layout()
plt.show()

fig, axes = plt.subplots(3, 5, figsize=(14, 8))
fig.suptitle("CIFAR-10 | Top: Original RGB Mid: Grayscale Bottom: Canny Ed",
for i in range(5):
    gray = cv2.cvtColor(cifar_x[i], cv2.COLOR_RGB2GRAY)
    edges = cv2.Canny(gray, 50, 150)
    axes[0, i].imshow(cifar_x[i])
    axes[0, i].set_title(f"Orig {i+1}")
    axes[0, i].axis('off')
    axes[1, i].imshow(gray, cmap='gray')
    axes[1, i].set_title(f"Gray {i+1}")
    axes[1, i].axis('off')
    axes[2, i].imshow(edges, cmap='gray')
    axes[2, i].set_title(f"Edges {i+1}")
    axes[2, i].axis('off')
```

```
plt.tight_layout()
plt.show()
```

MNIST | Top: Original Bottom: Canny Edges



CIFAR-10 | Top: Original RGB Mid: Grayscale Bottom: Canny Edges

